EX.NO:3 Exploring the different types of prompts- Straightforward Prompts - Tabular Format Prompting - Missing Word Prompting - Preceding Question Prompting.

1. Straightforward Prompts

Scenario 1: Artificial Intelligence

Prompt 1:
"Define Artificial Intelligence in one sentence."

Output:
Artificial Intelligence is the simulation of human intelligence in machines that are programmed to think and learn.

Prompt 2:
"List three real-world applications of Artificial Intelligence."

Output:

1. Virtual assistants (Siri, Alexa)
2. Self-driving cars
3. Medical diagnosis systems

Scenario 2: Cybersecurity

Prompt 1:
"Explain what a firewall is in simple terms."

Output:
A firewall is a security system that monitors and controls incoming and outgoing network traffic to protect a computer from unauthorized access.

Prompt 2:
"List three common types of cyber attacks."

Output:

1. Phishing
2. Malware
3. Denial-of-Service (DoS) attacks

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2. Tabular Format Prompting

Scenario 1: Programming Languages

Prompt 1:
"Provide a table comparing Python, Java, and C++ based on their programming paradigm, main features, and common applications."

Output:

Language	Paradigm	Main Features	Common Applications
Python	Object-Oriented, Procedural	Easy syntax, large libraries	Data science, web development
Java	Object-Oriented	Platform independent, secure	Enterprise software, Android apps
C++	Object-Oriented, Procedural	High performance, memory control	Game development, system software

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Scenario 2: Renewable Energy

Prompt 2:
"Provide a table listing five renewable energy sources and their major benefits."

Output:

Energy Source	Benefit
Solar Energy	Clean and unlimited energy from the sun
Wind Energy	Reduces dependence on fossil fuels
Hydropower	Produces large-scale electricity
Geothermal Energy	Reliable and constant energy source
Biomass Energy	Uses organic waste for power generation

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3. Preceding Question Prompting

Scenario 1: Blockchain Technology

Prompt 1:
"What is blockchain technology? Explain how blockchain ensures secure and transparent transactions."

Output:
Blockchain is a distributed digital ledger that records transactions across many computers. It ensures secure and transparent transactions by storing data in blocks that are linked together using cryptography. Once a transaction is recorded, it cannot be changed, making the system secure and trustworthy.

Scenario 2: Machine Learning

Prompt 2:
"Why is data important in machine learning? Describe how training data improves model accuracy."

Output:
Data is important in machine learning because algorithms learn patterns from the data they are trained on. Training data helps the model understand relationships between inputs and outputs, which improves the accuracy of predictions when the model encounters new data.

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4. Missing Word Prompting

Scenario 1: Computer Science

Prompt 1:
"The brain of the computer is called the ____."

Output:
CPU (Central Processing Unit)

Prompt 2:
"HTTP stands for HyperText ____ Protocol."

Output:
Transfer

Scenario 2: Environmental Science

Prompt 1:
"Plants release ____ gas during photosynthesis."

Output:
Oxygen

Prompt 2:
"The layer that protects Earth from harmful UV radiation is called the ____ layer."

Output:
Ozone